McMaster University | Engineering

Proxi - A Dynamic AI Agent

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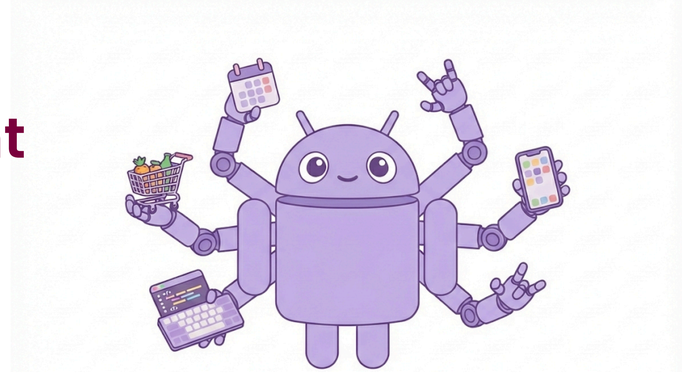


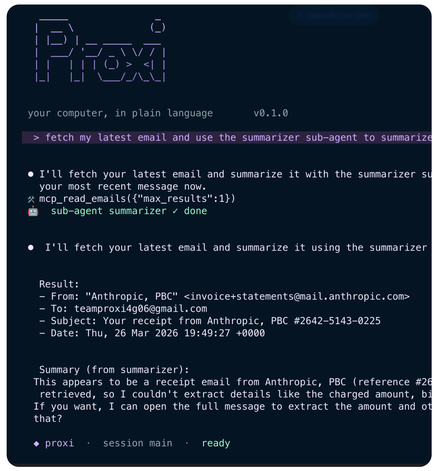
Introduction

Proxi is a modular assistant platform designed to connect users, tools, and language models within a single workspace focused system. It combines a CLI/TUI experience with a gateway based architecture, enabling reliable interaction across agents, memory, safety controls, and external integrations such as MCP tools. By emphasizing security, observability, and extensibility, Proxi supports practical automation workflows while remaining maintainable for long term development and team collaboration.

Objectives

1. Build an accessible assistant that helps users interact with digital tools more easily.
2. Provide a reliable command and response workflow across core tasks.
3. Design Proxi to be modular and extensible so new tools/features can be added over time.
4. Prioritize usability, clarity, and a smooth user experience for different user groups.
5. Ensure the system is secure, stable, and maintainable for future development.
6. Support speech interaction as an available feature, while keeping the project scope realistic for the current phase.

Subjects and Methods



Subjects

- Primary users: individuals who need more accessible computer interaction
- Includes: people with disabilities, older adults, and users with limited digital literacy
- Secondary users: caregivers, educators, and family members who support primary users

Methods

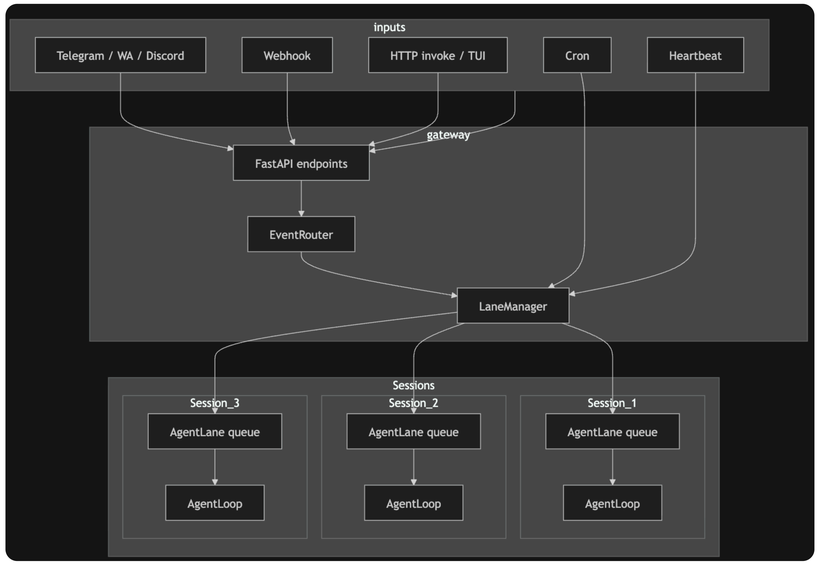
- User centered development approach
- Task based usability testing with representative tasks
- Observe task completion, usability barriers, and interaction flow
- Collect user feedback through short surveys
- Measure ease of use, clarity, and overall satisfaction
- Iteratively improve Proxi based on testing results to enhance accessibility and reliability

Results

- Proxi achieved its functional objectives and handled major workflows reliably in testing.
- Users were able to complete most common tasks through the high level interface after a short onboarding period.
- System responses were generally timely and clear, supporting a smooth interaction flow.
- The platform remained stable during extended use and handled invalid or incomplete inputs safely.
- Safety and accountability features, including confirmation for sensitive actions and logging, worked as expected.
- The main improvement areas are clearer phrasing support for ambiguous commands.

Challenges

- Balancing project ambition with a realistic capstone scope and timeline
- Defining objectives that were both general (solution-neutral) and still measurable
- Handling ambiguous follow-up commands and improving clarification behavior
- Managing integration edge cases across tools/services (e.g., loading/state mismatches)
- Refining usability and voice interaction reliability without overextending final scope
- Coordinating team workload and maintaining consistent documentation/testing quality across deliverables



proxi

Features

- Accessible high-level assistant for everyday tasks
- Text + speech input support (speech in progress)
- Natural-language intent understanding
- Agent/tool-based task planning and execution
- Common workflow support: email, calendar, browser, files, workspace
- Clear text feedback with optional spoken-style flow
- Short-term conversational context for follow-ups
- Built-in undo, cancel, and help
- Confirmation for privileged/destructive actions
- Interaction logging for traceability/debugging
- Modular, extensible architecture
- Interactive TUI + one-shot CLI modes
- TUI QoL: command palette, plan/todo views, collaborative forms, input history

References

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